MILITARY SPECIFICATION

WELDING MACHINES, ELECTRICAL RESISTANCE, SPOT, ROCKER ARM, AIR OPERATED, THREE PHASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three phase, air operated, rocker arm, electrical resistance, spot welding machines with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following types, classes and sizes. The type, class, and size to be furnished shall be as specified (see 6.2.1).

Type I - Low frequency converter.
Type II - Rectifier direct current.

Class A - 24 inch throat depth.
Class B - 36 inch throat depth.
Class C - 48 inch throat depth.
Class D - 60 inch throat depth.
Size 0 - 50 KVA rating at 50 percent duty cycle.
Size 1 - 75 KVA rating at 50 percent duty cycle.
Size 2 - 100 KVA rating at 50 percent duty cycle.
Size 3 - 125 KVA rating at 50 percent duty cycle.
Size 4 - 150 KVA rating at 50 percent duty cycle.
Size 5 - 200 KVA rating at 50 percent duty cycle.
Size 6 - 250 KVA rating at 50 percent duty cycle.
Size 7 - 300 KVA rating at 50 percent duty cycle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Throat depth. The welding machine throat depth is the unobstructed distance, measured in the welding plane, from the centerline of the welding electrodes to the first obstruction of the machine frame.

1.2.2 Machine size. The machine size is expressed in terms of kilovolt amperes (KVA) at 50 percent duty cycle of the welding transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-6858 - Welding, Resistance, Spot and Seam.

MIL-W-45223 - Welding, Spot, Hardenable Steels.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models capable of producing spot welds on ferrous and nonferrous materials in accordance with MIL-W-6858, MIL-W-45223, and the requirements of this specification. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements contained in 29 CFR, Chapter XVII, Part 1910. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

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3.2.8 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or welding machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

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3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

3.4 Components. The welding machine shall consist of not less than a frame and base, an enclosure, an electrical system, a arm and horn assembly, a pneumatic system, and a water cooling system.

3.4.1 Frame and base. The frame and base of the welding machine shall be an integral unit of steel weldment box construction, ribbed to minimize deflection under load, with provisions for hoisting. The base shall provide for floor mounting. Access shall be provided to permit inspection and servicing of the equipment installed within the frame. The frame shall be designed to permit positioning of the lower arm or knee and lower horn assembly.

3.4.2 Enclosure. The enclosure metal shall be corrosion-resistant or treated to resist corrosion. Ventilation and cooling shall be adequate to prevent temperature in excess of design requirements, under normal operating conditions of rated power output.

3.4.3 Electrical system. The electrical system shall include the welding transformer, rectifiers, timers, meters, circuit protection devices, disconnect switches, and all controls required for operation. The electrical system and components shall conform to the applicable requirements of NFPA 70. Each subpanel shall be arranged so that it may be removed independently of any other subpanel. All terminals and units shall be clearly identifiable from the machine's electrical wiring diagram. Adequate ventilation of the control cabinet shall be provided. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.3.1 Electric service requirements. The welding machine shall operate on a three phase, 60 hertz, primary electric service at any of the following voltages; 208, 230, 460, or 575. Unless otherwise specified (see 6.2.1), the welding machine shall be initially wired for operation on 460 volts.

3.4.3.2 Main disconnect switch. The welding machine shall have a main disconnect switch or circuit breaker. Unless otherwise specified, a circuit breaker shall be furnished and shall be mounted on the machine panel. When specified (see 6.2.1), a main disconnect switch shall be furnished. A safety interlock or circuit breaker shall provide protection to the operator when any access door is opened to an area that presents a hazard.

3.4.3.3 Solid state components. Solid state design shall be used throughout for electronic components. Use of relays shall be kept to an absolute minimum with only such relays as required to ensure adequate safety and reliable operation will be used. Relays shall be the plug in type where practical.

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3.4.3.4 Welding transformer. The welding transformer shall be mounted in the welding machine. The transformer shall be either the low frequency conversion type or the rectifier type output. The transformer shall be operated on an essentially balanced three phase power supply at a vector power factor of not less than 85 percent. The welding transformer KVA rating and secondary current rating shall be in accordance with table I for the machine size specified. The transformer shall be water cooled and thermostatically controlled to minimize condensation. A temperature responsive device shall be provided to prevent energizing the transformer if the temperature of the transformer exceeds design limits. All connecting surfaces of the secondary circuit shall be silver plated. The transformer insulation shall conform to RWMA Bulletin Number 16, class B, or better. When specified (see 6.2.1), the transformer shall have a series/parallel switch in the primary circuit.

3.4.3.5 Secondary current rectifier system. The secondary current rectifier system for the type II welding machine shall be the silicon diode type using three phase, full wave circuit arrangement. The forward current rating and reverse peak voltage rating of the diodes shall be compatible with the size machine specified. The silicon diodes shall be mounted on a water cooled conductor. Within each phase the impedance of each diode circuit shall be similar allowing each diode to load equally. The diodes shall be arranged for ease of examination, installation, and removal. Surge suppression protection shall be provided. The rectifier system shall have a protective control circuit which will open the line circuit breaker upon excessive temperature.

3.4.3.6 Electric controls. The electric controls shall consist of the welding power control, timing and sequence controls, a welding contactor, current adjustment control, and auxiliary and accessory controls. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms.

3.4.3.6.1 Operator control panel. The operator control panel shall have an "on-off" or "weld-no weld" switch, a "power on" and sequence lights, control function switches, and any other controls or switches required to meet the requirements specified herein.

3.4.3.6.2 Welding power control unit. The machine shall have a solid state synchronous precision, three phase welding power control unit. The control unit may be located either in a separate cabinet or mounted as an integral part of the machine.

3.4.3.6.3 Welding contactor. The welding contactor shall be a silicon controlled rectifier (SCR) type. The SCR contactor shall be water cooled and thermostatically protected. Means shall be provided to prevent welding if the water is shut off or if the water flow is restricted. The SCR contactor shall be provided with over current and over voltage protection. The

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TABLE I. Machine size and capacities.

Machine size	KVA rating	Type I machines				Type II machines			
		Secondary short circuit current (amps)		Electrode force range (lbs.) 1/		Secondary current range (amps)		Electrode force range (lbs.) 1/	
		min.	max.	min.	max.	min.	max.	min.	max.
0	50	5,000	50,000	150	1,000	3,500	35,000	150	1,000
1	75	6,000	60,000	200	2,900	6,000	60,000	300	2,900
2	100	7,000	70,000	350	4,300	6,500	65,000	450	4,300
3	125	8,000	80,000	450	5,900	6,700	67,000	550	4,900
4	150	9,000	90,000	600	6,800	7,000	70,000	600	5,000
5	200	13,000	130,000	800	10,200	8,000	80,000	600	5,600
6	250	15,000	150,000	1,800	18,000	10,000	100,000	1,200	12,000
7	300	17,000	170,000	1,900	19,000	11,000	110,000	1,500	15,000

1/ With the pneumatic system being supplied with an air pressure of not less than 80 PSI.

NOTE: When a value range is shown the requirements shall be not more than the minimum and not less than the maximum value shown.

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welding contactor and rectifier may be a single unit operating in the primary circuit of the welding transformer of the type I machines, or they may be separate units operating in the primary and secondary circuits, respectively, of the type II machines. The firing circuit of each SCR bank shall be provided with a fuse or other protective device and a light to indicate when the firing circuit has been interrupted by the protective device. On low frequency converter systems, the welding current output of the positive and negative rectifier shall be inherently balanced.

3.4.3.6.4 Timing and sequencing controls. The control equipment shall provide operating control and adjustment of the welding variables so as to ensure that optimum values of force and current timing and sequencing can be repetitively attained to produce spot welds in accordance with MIL-W-6858. The welding sequence shall be of the non-beat type. The control shall be of the sychronous precision type. The various timing and sequencing functions shall be grouped on a plug-in chassis. Provisions shall be made for the removal or addition of extra functions. The controls shall have not less than the following standard, or equivalent timing functions; squeeze, weld, cool, post heat, forge, hold, and off. Current decay, or inversion, shall be furnished on the type I machine. When specified (see 6.2.1), preheat, weld, and temper functions shall be provided for spot welding hardenable steel and low alloy steels in accordance with MIL-W-45223. The timing control input shall be calibrated in cycles, based on the electrical service line frequency of 60 Hz, except that the preheat, weld, and temper interval control input may be calibrated in either cycles or impulses. The timing controls shall have a minimum range of 0 through 99 cycles or impulses. The forge function shall be programmable and shall have a minimum of 0.1 to 999.9 cycles with a repetitive accuracy of ±.05 cycle. The controls shall have signal lights on the panel to indicate the operating sequence of the welding cycles and to aid in trouble shooting when a malfunction occurs. The timing and sequencing controls shall be selected and adjusted by means of knobs, switches, or digital input.

3.4.3.6.5 Current adjustment control. The welding current shall be continuously variable from the lower to the upper limits of the machine capacity. The current adjustment may be by means of phase shift heat control, switches, digital input selection, or a combination of any of these. The control shall be graduated in increments of 1 from 0 to not less than 99 or 10 percent to not less than 99 percent for convenience in selecting settings. The current adjustment maximum setting shall product maximum short circuit current on type I machines or maximum welding secondary current on type II machines. The preheat, weld, and temper current controls shall be independently adjustable when specified (see 3.4.3.6.4).

3.4.3.6.6 Automatic voltage compensator. The welding machine shall have an automatic voltage compensator (AVC) system to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ±2 percent for a line voltage variation of ±10 percent. The AVC shall employ three phase line voltage sensing. Full compensation shall

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occur within 1/3 Hz for any sudden change in line voltage in any of the three phases. The AVC shall operate on the preheat, weld, and temper currents when specified (see 3.4.3.6.4).

3.4.3.6.7 Control circuit voltage. Control equipment shall be capable of operating on control circuit voltage which varies as much as +10 percent from the adjusted voltage. All switch inputs shall be isolated from the input electric power by a low voltage transformer having a secondary voltage no greater than 24 volts.

3.4.3.6.8 Accessory controls. Accessory controls shall consist of not less than the following:

3.4.3.6.8.1 Foot switch. The foot switch shall control the upper rocker arm assembly and its associated pneumatic system and initiate the welding cycle. A two stage foot switch shall be supplied to initiate the rocker arm action and apply electrode force in the first stage and the welding sequence in the second stage.

3.4.3.6.8.2 Tip dress control. The tip dress control shall operate to lower the rocker arm under a reduced force for cleaning and aligning the electrodes.

3.4.3.6.8.3 Contact pressure gauge switch. The contact pressure gauge shall be a presettable pressure switch that shall prevent continuation of the welding sequence, as initiated by the foot switch, until the air pressure in the pneumatic head is equal to the contact gauge setting.

3.4.3.6.8.4 Electrode force control. The electrode force control shall permit selection of a constant high, constant low, or a variable force sequence.

3.4.3.6.8.5 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle. The weld shall terminate immediately when the retraction switch is released and air pressure is lost.

3.4.3.6.8.6 Weld/no weld or on/off switch. A weld/no weld switch shall be provided. In the "no weld" or "off" position, the control sequence shall actuate with no welding current flow.

3.4.3.6.8.7 Repeat on/off or repeat/non-repeat switch. A repeat on/off switch shall be provided. In the "off" or "non-repeat" position only one welding cycle will occur. In the "on" or "repeat" position the welding cycle will repeat automatically after a controlled off time as long as the foot switch is energized.

3.4.3.6.8.8 One, two, or three phase welding switches. On type I low-frequency converter machines individual switches shall be provided to select and weld with unidirectional or bidirectional welding current of one,

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two, or three phases. Lights or readouts shall indicate firing of individual SCR banks.

3.4.4 Arm and horn assemblies. The arm and horn assemblies shall be designed and constructed to maintain original alignment and to minimize deflection. With maximum rated electrode force applied, and the upper and lower arms unsupported by any bracing, the horizontal deflection of the upper and lower electrodes shall not exceed the value specified in table II. The vertical deflection of the lower electrode holder shall be not more than the value specified in table II. The horns shall be made of solid copper and have a diameter as specified in table III. Each horn shall be adjustable in and out at least three inches from nominal throat depth. The horns shall have clamps that will not mar the electrode holders after repeated adjustments.

3.4.4.1 Rocker arm. The rocker arm shall fulcrum on anti-friction bearings that are protected against dust and foreign matter. The rocker arm movement shall be operated by an air cylinder that is actuated by the foot switch. The rocker arm electrode force shall be adjustable and calibrated for its specified throat depth. The rocker arm shall be electrically connected to the secondary of the welding transformer by a flexible shunt having silver plated connections. The working stroke and total electrode opening shall be as shown in table III.

3.4.4.2 Lower arm or knee. The lower arm shall be designed and mounted on the frame of the welding machine to provide a vertical adjustment of the horn spacing of not less than that specified in table III. The knee shall have a socket to permit motion of the horn and alignment of the lower electrode holder in reference to the upper electrode. A hydraulic jack shall be permanently mounted on the base of the machine to aid in the vertical adjustment of the arm.

3.4.4.3 Electrodes and electrode holders. The electrodes and electrode holders shall conform to the requirements of table III and to RWMA Bulletin Number 16. The quantity, type, size, and alloy class, as applicable, of electrodes, electrode holders, and adapters required shall be as specified (see 6.2.1).

3.4.5 Pneumatic system. The pneumatic system and components shall conform to the requirement of RWMA Bulletin Number 16 and this specification.

3.4.5.1 Pneumatic system operation. The pneumatic system air cylinder shall be of the double action, adjustable stroke type, and shall be operated with not less than 80 pounds per square inch (psi) input air service. The inlet connection shall be marked for the pressure requirements. The pneumatic system shall include valving, piping, regulators, gauges, lubricators, moisture drain, wiring, switches, and all related equipment necessary to permit use of the electrode force modes, as follows; constant low,

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TABLE II. Maximum permissible electrode limits.

Machine size	Throat depth (inches)	Vertical deflection of the lower electrode holder (inches)	Horizontal deflection difference between upper and lower electrodes (inches)
0 and 1	24	.0286	.0036
	36	.0330	.0036
	48	.0352	.0036
	60	.0374	.0048
2	24	.0308	.0036
	36	.0330	.0048
	48	.0352	.0048
	60	.0374	.0060
3 and 4	24	.0319	.0048
	36	.0352	.0060
	48	.0396	.0060
	60	.0418	.0060
5	24	.0330	.0048
	36	.0352	.0060
	48	.0418	.0060
	60	.0440	.0060
6	24	.0341	.0048
	36	.0385	.0060
	48	.0440	.0060
	60	.0495	.0060
7	24	.0341	.0060
	36	.0385	.0072
	48	.0451	.0072
	60	.0528	.0072

NOTE: Deflection requirements are not more than value shown.

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TABLE III. Electrode, horn, and holder dimensions.

Machine size	Electrode opening (inches) working	Electrode opening (inches) total	Horn spacing (inches)	Horn diameter (inches) min.	Horn diameter (inches) max.	Electrode holder diameter (inches)	Electrode holder taper
0	1/2	4	6	3	3-1/2	1-1/4	RWMA #5 (#2 Morse)
1	1/2	4	6	3	3-1/2	1-1/2	RWMA #5 (#2 Morse)
2	1/2	4	7	3	3-1/2	1-1/2	RWMA #7 (#3 Morse)
3	1/2	4	7	3	3-1/2	1-1/2	7/8 inch 8 degree included angle or 7/8 inch threaded
4	1/2	5	8	3 1/2	4	1-1/2	1 inch 8 degree included angle or 1 inch threaded
5	3/4	5	8	4	5	1-3/4	2 inch threaded
6	1	5	10	5	6	1-3/4	2 inch threaded
7	1	5	12	5	6	2	2-1/2 inch threaded

NOTE: Requirements are not less than the values shown. When a range is shown the requirements shall be not more than the minimum and not less than the maximum value shown.

constant high, or variable. Air cushions or air speed valves shall be provided to ensure maximum shockless operation.

3.4.5.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches and shall be calibrated in psi or pounds of electrode force. Gauge accuracy shall be within ±0.5 percent of the maximum reading of the gauge. Means of shock protection shall be provided to prevent gauge damage.

3.4.5.3 Air regulators. The air regulators shall be self bleeding. Regulators and solenoid valves shall be installed with unions. The regulators shall maintain the pressure ±0.5 psi within a 2 through 100 psi range as long as the supply pressure is 2 psi higher than the preset regulator pressure. The rise of the electrode force, from its weld force to its forge force level, shall be 75 percent accomplished in no more than 10 milliseconds. The air system for the upper rocker arm shall produce the required electrode force to perform the welds specified herein.

3.4.6 Water cooling system. The welding machine shall have a water cooling system that will prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and

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flow requirements. The system shall have parallel flows from the manifold to provide proper cooling of the welding transformer, the secondary winding and circuit conductors, the primary welding current contactors, and the upper and lower electrodes. The piping in the system shall be corrosion resistant to tap water services. The system shall be provided with a single outlet spilling into an open cup drain for a visual check on the water flow. The water flow lines to the electrodes shall have solenoid valves to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall be provided with water saving devices, an inlet strainer, and a closed manifold drain system with flow indicators.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than those specified in tables I, II, and III for the machine size specified.

3.6 Performance. The welding machine shall be capable of producing class A spot welding conforming to the requirements of MIL-W-6858, table IV, and when specified MIL-W-45223.

TABLE IV. Machine welding capacity two pieces of equal thickness.

Machine size	Spots per minute	Aluminum and magnesium alloys (inches)		Austenitic steel and 300 series (inches)		Nickel cobalt alloys (inches)		SAE 4130 or titanium TIV-GA1-4V (inches)	
		min.	max.	min.	max.	min.	max.	min.	max.
0	20	.012	.025	.012	.040	.012	.025	.010	.032
1	16	.016	.064	.016	.070	.015	.060	.014	.067
2	14	.020	.081	.022	.109	.018	.075	.018	.090
3	12	.025	.091	.025	.115	.020	.080	.025	.095
4	12	.032	.102	.030	.125	.027	.093	.027	.105
5	10	.040	.125	.048	.150	.042	.120	.042	.135
6	8	.064	.156	.060	.203	.048	.164	.048	.185
7	7	.081	.166	.080	.223	.058	.188	.064	.200

NOTE: Requirements are not less than the value shown. When a range is shown the requirements shall be not more than the minimum and not less than the maximum value shown.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial welding machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

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3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, stenciling, silk screening, or by any other method suitable for the use intended. Markings shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below and that required in RWMA Bulletin Numer 16, Name and Rating data plates. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's model designation
Date of manufacture
Contract Number or Order Number
National Stock Number or Plant Equipment Code

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the welding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the welding machine. The meter shall operate whenever the weld switch is turned on. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not more than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

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3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

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4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be connected to the specified source of electric, air, and water supply in accordance with the manufacturer's instructions. The welding machine shall be operated without welding for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance welding test. The welding machine shall be tested for welding capability by performing the tests in MIL-W-6858 for class A welds. The test materials shall be 2024T3 or 7075T6 alclad aluminum alloy and stainless steel type 203 or 304. Tests shall be made using each of these materials in the maximum and minimum thicknesses specified in table IV according to the machine size specified. When specified (see 3.4.3.6. 4), the machine shall be tested for capability of welding hardenable materials in accordance with MIL-W-45223, by welding a low alloy steel, type SAE 4130, of the maximum thickness specified in table IV according to the machine size specified. If the electrodes arc upon separation in any of the above tests the machine shall be considered as having failed the test in which arcing occured.

4.7.3 Electrode deflection test. The welding machine shall be tested to determine the deflection of the upper and lower electrodes. The electrodes shall be positioned at the nominal throat depth location with the pneumatic system adjusted to provide maximum electrode force as specified in table I. Without the lower arm or horn being supported by a brace or any other device that can serve as a brace, the electrodes shall be brought to bear upon a plate and roller assembly mounted similar to figure 1. The diameter of the rollers and the thickness of the plates shall be not less than one-half the electrodes diameter and preferrably not more than the value. The difference between the upper electrode and lower electrode horizontal deflection, and the vertical deflection of the lower electrode holder shall be not more than that specified in table II.

4.7.4 Automatic voltage compensator test. The performance of the automatic voltage compensator shall be measured by means of a calibrated recording instrument. With the welding or secondary current adjustment set first at 50 percent and then at 90 percent of maximum, the secondary current and

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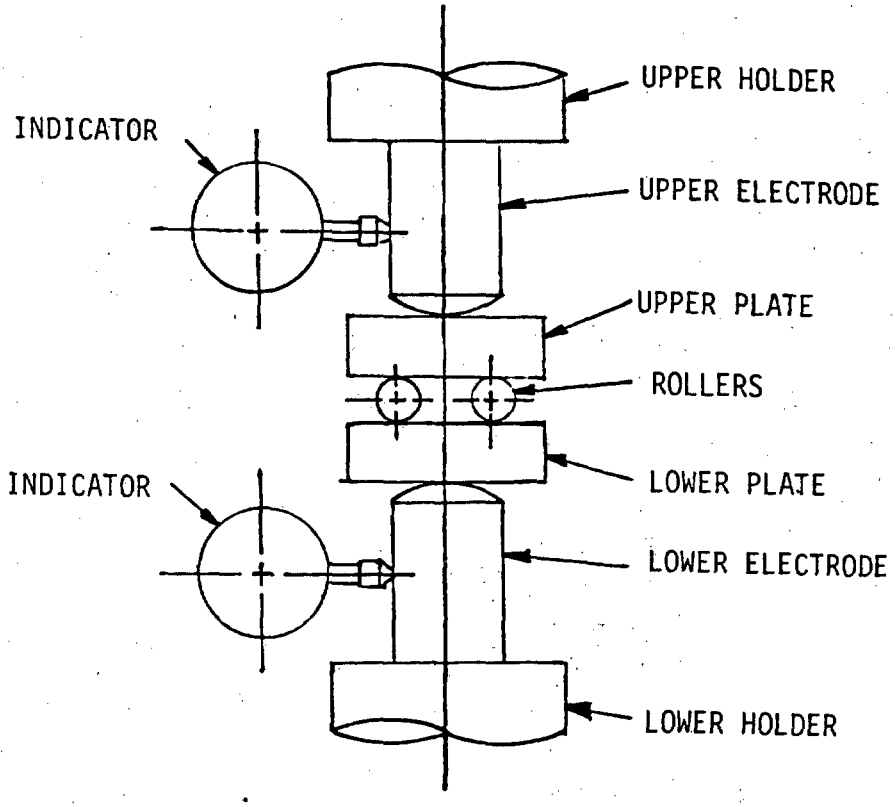


FIGURE 1. DETAIL OF DIAL INDICATOR SET-UP FOR MEASURING ELECTRODES HORIZONTAL DEFLECTION.

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primary voltage shall be recorded as ±10 percent step changes are introduced into the primary voltage while welding or secondary current is flowing. Alternately, the compensation and response time may be tested by causing the voltage change to occur only on the control of the compensator, in which case, the 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ±2 percent. The secondary current compensation and response time shall be as specified in 3.4.3.6.6.

4.7.5 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin Number 16.

4.7.6 Forge force rise time. With the machine sequence set in the variable force mode and the machine controls adjusted to the proper values for welding two pieces of the maximum thickness of alclad aluminum specified in table IV in accordance with MIL-W-6858 the rise time of the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The rise of the electrode force from its weld force level to its forge force level shall be 75 percent accomplished in no more than the time specified in 3.4.5.3.

4.7.7 Rated capacity test. The welding control repeat switch shall be placed in the "on" position and the machine controls adjusted and made to operate as for welding of the maximum thickness aluminum specified in table IV for the size machine being tested. The test shall be conducted without any material between the electrode tips and the current control adjusted so that the secondary current under short-circuit condition is equal to the current used for welding the aluminum. With the off-time set so that the number of welding operations per minute is not less than that specified in table IV, the equipment shall operate for a total of not less than 2 hours without making any adjustment of the controls or relays. During this test, the secondary current flow time and amplitude shall be measured by means of a calibrated recording instrument for 3 successive periods of flow at the beginning of each hour and at the end of the test. The variation in the peak value of the welding current shall not exceed 5 percent in any one set of periods, and shall not exceed 10 percent over the entire tests. A maximum of one shut down period, not to exceed 5 minutes shall be allowed to dress the electrodes, if necessary. If the electrodes arc upon separation in any one operation, the equipment shall be considered as having failed the test in which arcing occurred. At the end of the 2 hour period the equipment shall be shut down and all parts examined for heating. The temperature limits specified in RWMA Bulletin Number 16 shall not be exceeded. The contacts of the relays, if any, shall be examined, and if there is evidence that arcing appears to be excessive, the supplier shall be required to make changes which will reduce the arcing. If the equipment fails to conform to any of these requirements, or fails to operate properly during the test, the test shall be discontinued, and the supplier permitted to make any necessary adjustments. After adjustment, the test in which the equipment failed shall

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be completely rerun. If the equipment fails to pass the rerun test, failure shall be cause for rejection.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The welding machines covered by this specification are intended for use in producing class A spot welds in accordance with this specification and the requirements of MIL-W-6858 and MIL-W-45223 as applicable, under production conditions. The welding machine is intended for broad and critical spot welding requirements of the aircraft, missile, and spacecraft manufacturing industry, including welding of critically loaded primary structural assemblies of heat resistance alloys and exotic metals.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If welding machine design and construction is required in a specific dimensional system (US or SI), state required system (3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Electrical input power, if different (see 3.4.3.1).

i. Circuit breaker and its mounting, if different (see 3.4.3.2).

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j. Series/parallel switch, if required (see 3.4.3.4).

k. Control functions for welding hardenable and low alloy steels in accordance with MIL-W-45523, if required (see 3.4.3.6.4).

l. Specify quantity, size, type, and alloy class of electrodes, electrode holders, and adapters required (see 3.4.4.3).

m. Water saving devices, if required (see 3.4.6).

n. Size and capacity, if different (see 3.5).

o. Additional performance features, if required (see 3.6.1).

p. Standard equipment, if different (see 3.7).

q. Specify and fully describe optional equipment, if required (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

s. Lubrication chart or plate, if different (see 3.10.1).

t. Nameplate, if different (see 3.10.2).

u. Hour meter, if different (see 3.11).

v. First article inspection, if different (see 4.4).

w. Quality conformance inspection, if different (see 4.5).

x. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the welding machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machine in the environment. The procuring activity is advised to analyze the existing hazard levels, in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (f).

6.5 Cross-reference of classification changes. The welding machine classification changes from the previous issue of this specification are as follows:

MIL-W-80087B	MIL-W-80087C
Type I	Type I
Type II	Type II
24 inch throat depth	Class A (New class)
36 inch throat depth	Class B (New class)
48 inch throat depth	Class C (New class)
60 inch throat depth	Class D (New class)
Size 0	Size 0
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	Size 4
Size 5	Size 5
Size 6	Size 6
Size 7	Size 7

6.6 Subject (key word) listing.

Automatic voltage compensator
Electrode deflection
Electrode forge force
Resistance welding standards
Sequence timer
Synchronous welding power control
Throat depth
Tip dress control
Welder, resistance
Welder, spot

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Welding contactor
Welding electrodes
Welding machine, resistance, spot, rocker arm type
Welding transformer

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA- GS

User activities:
Army - AV, SM
Navy - OS

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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